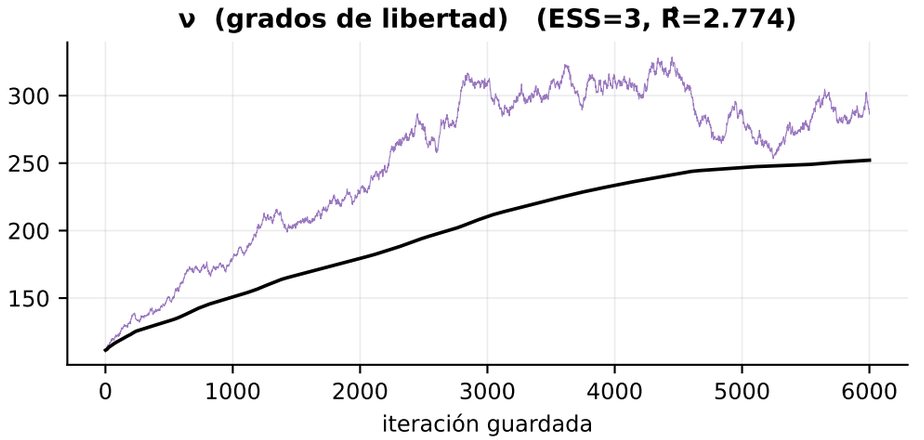
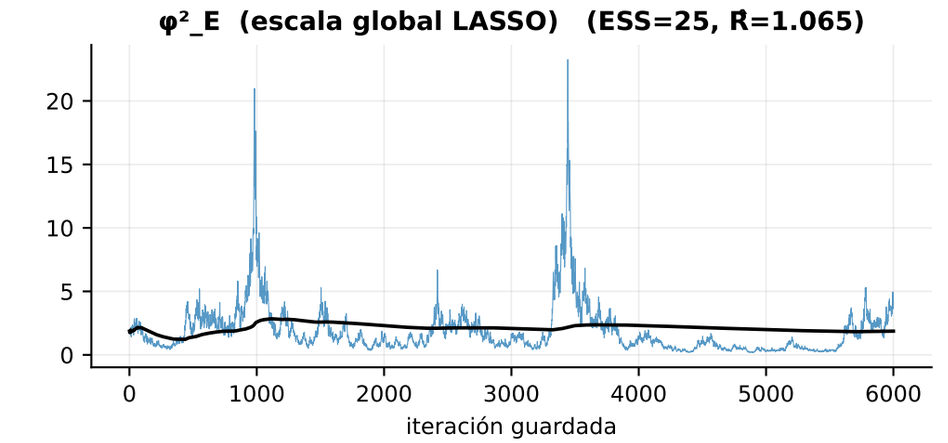
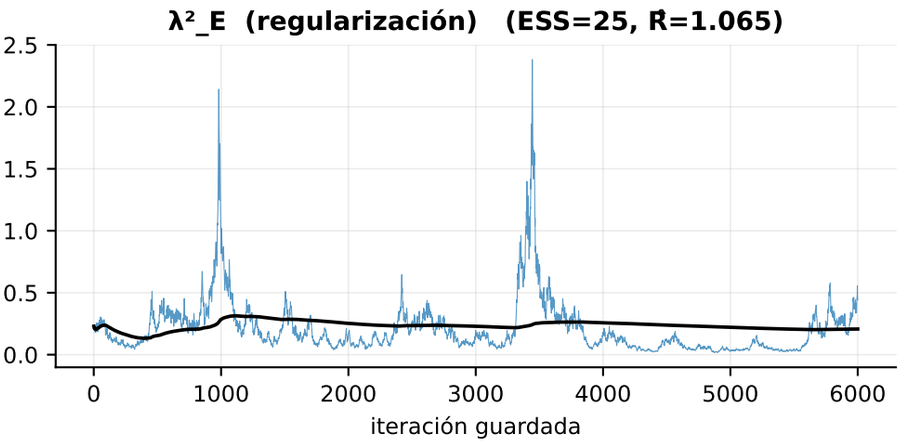
M4 — Convergencia I: trazas de los parámetros globales

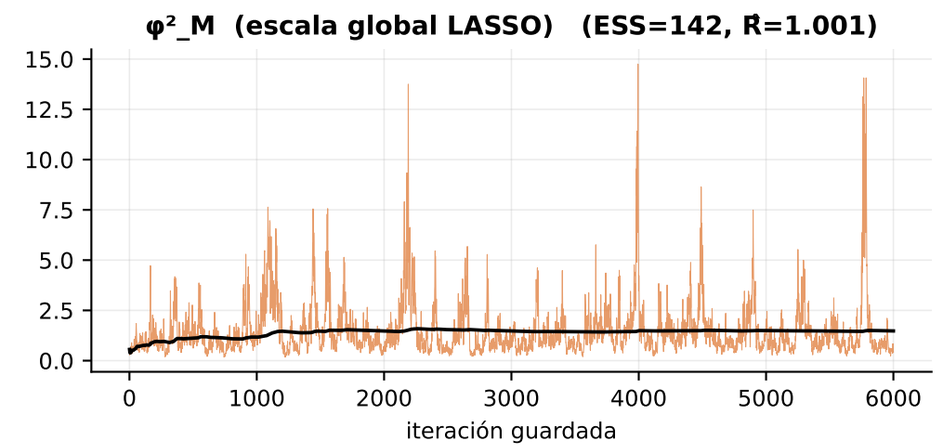


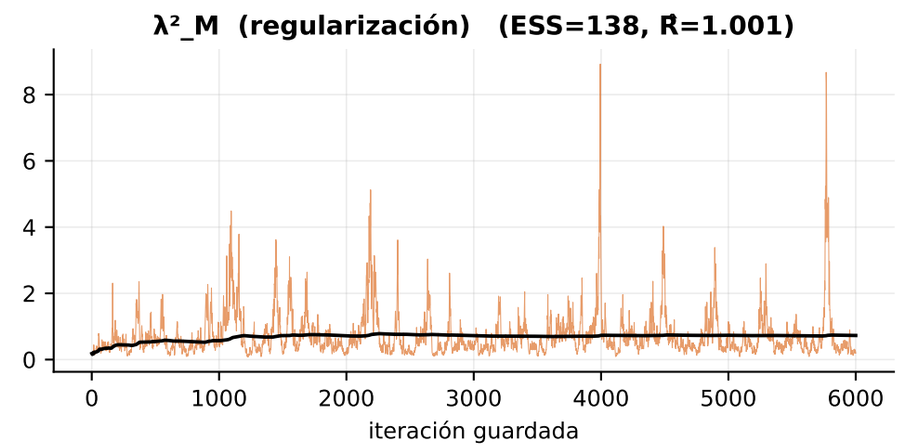
Trazas posteriores + media acumulada (línea negra)
Cadena post burn-in del muestreador híbrido M4.

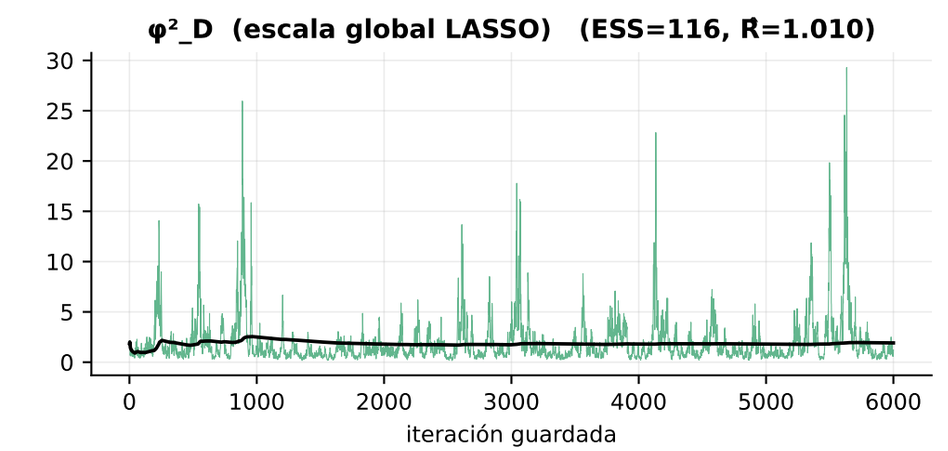
S = 6000 guardadas (burn-in=5000, thin=5)
Aceptación MH de ν = 0.329

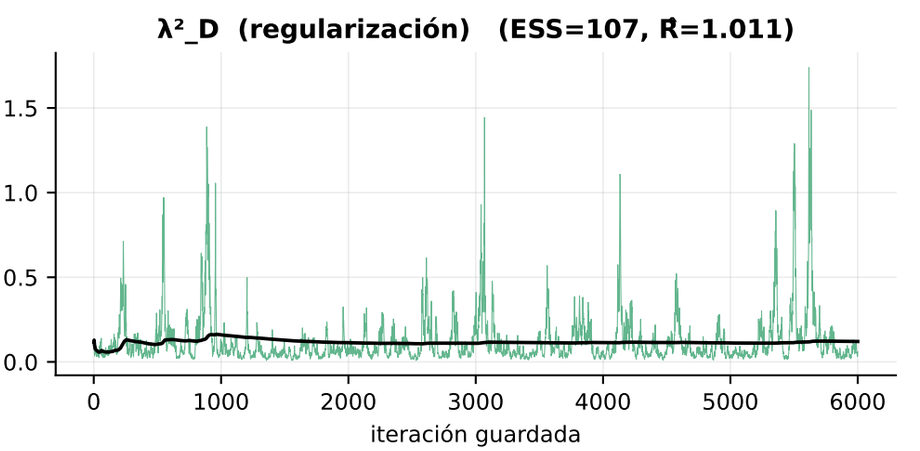




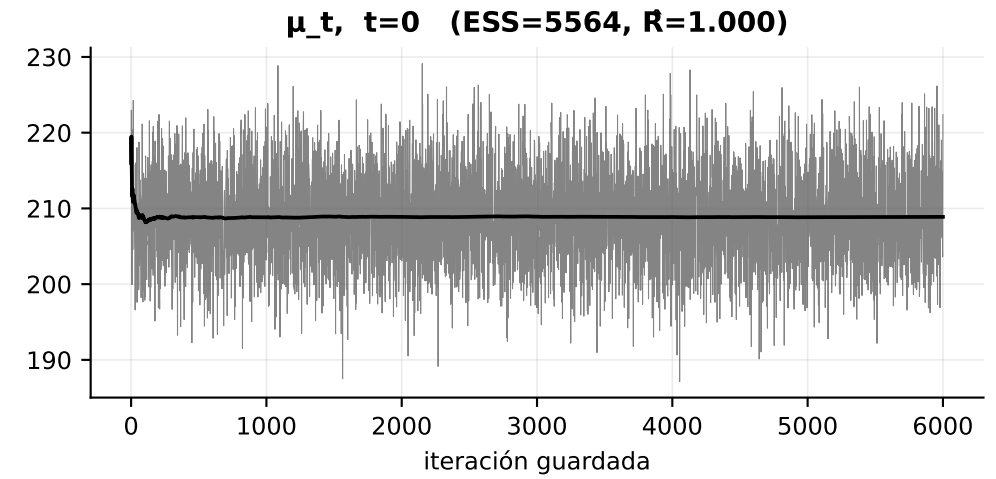


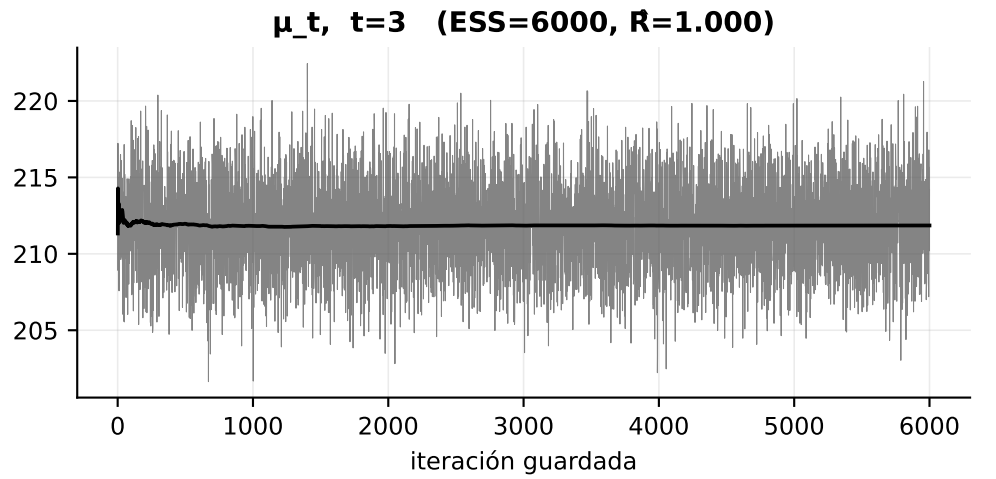


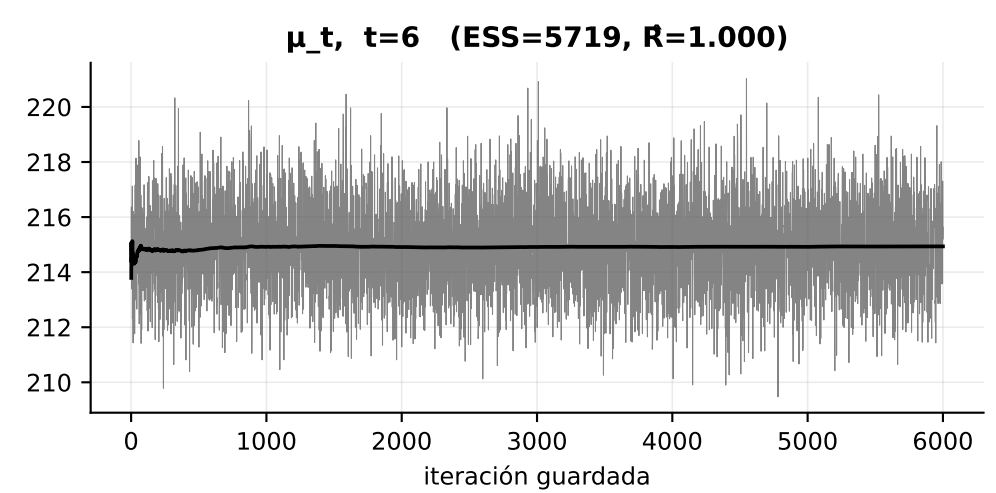


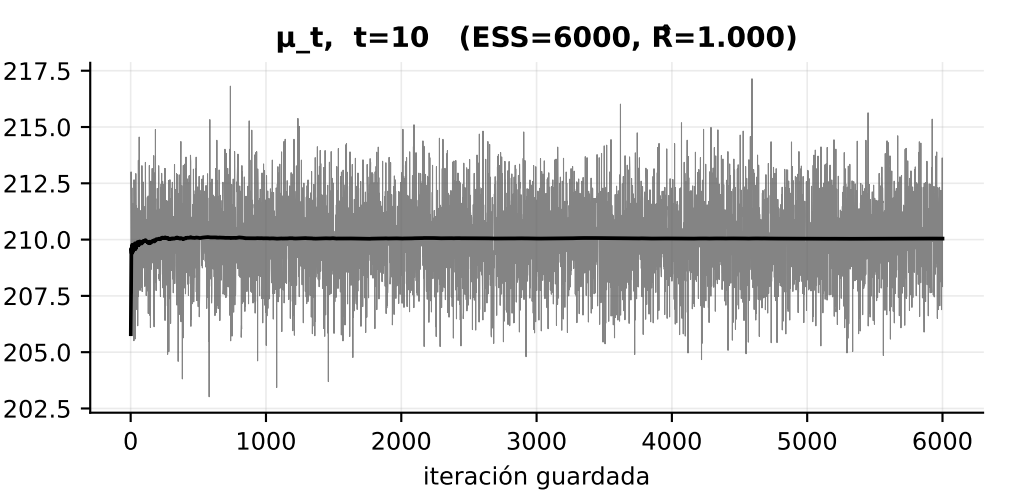


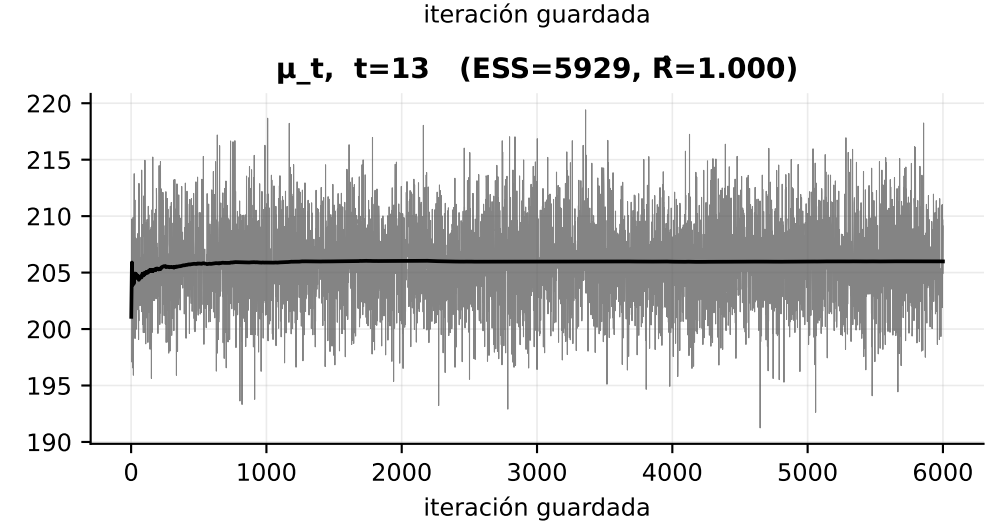
M4 — Convergencia II: trazas del intercepto dinámico μ_t

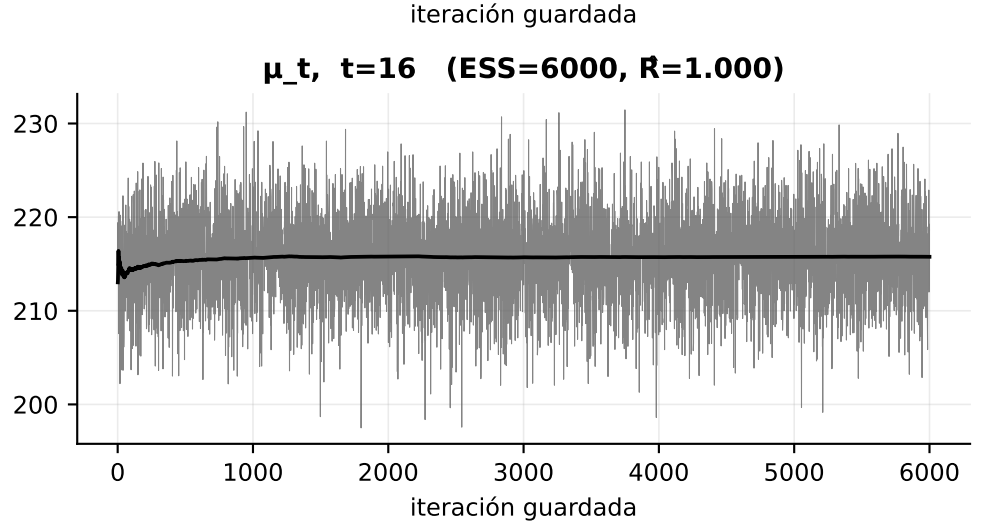




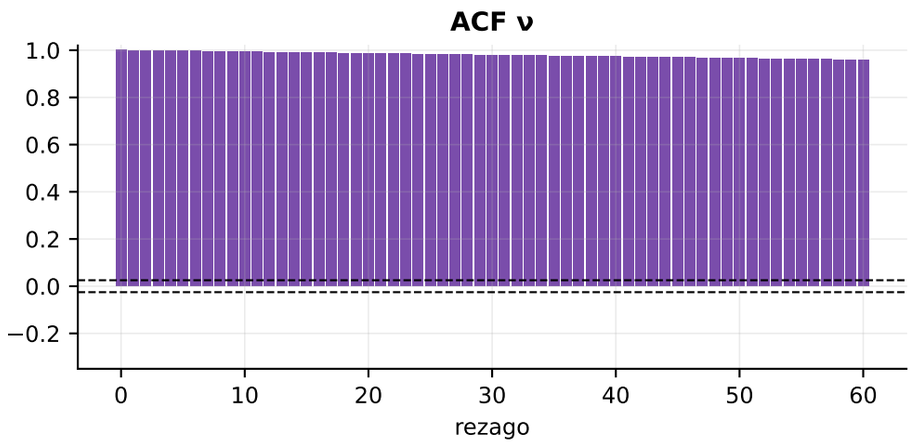


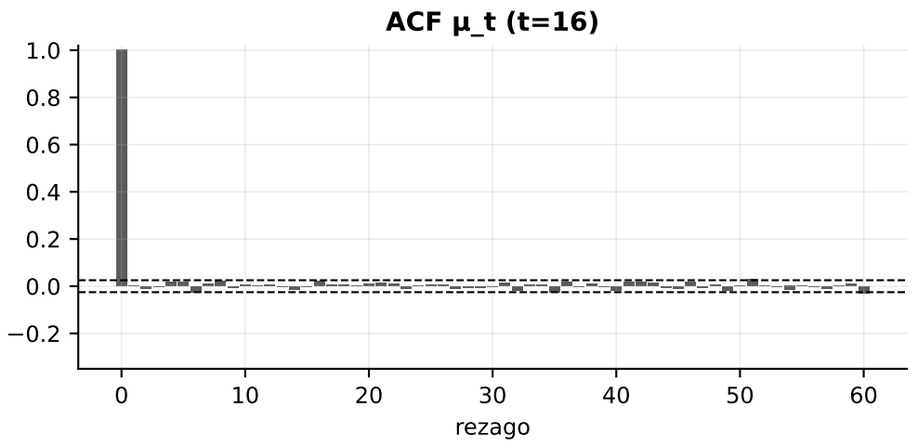


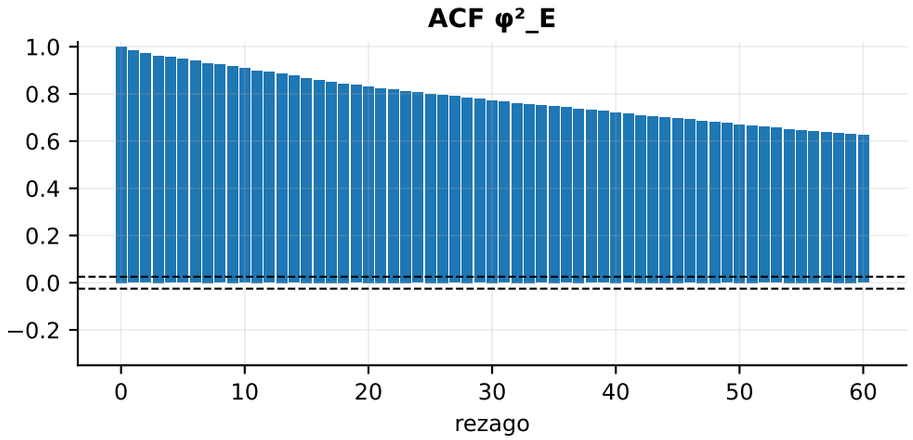


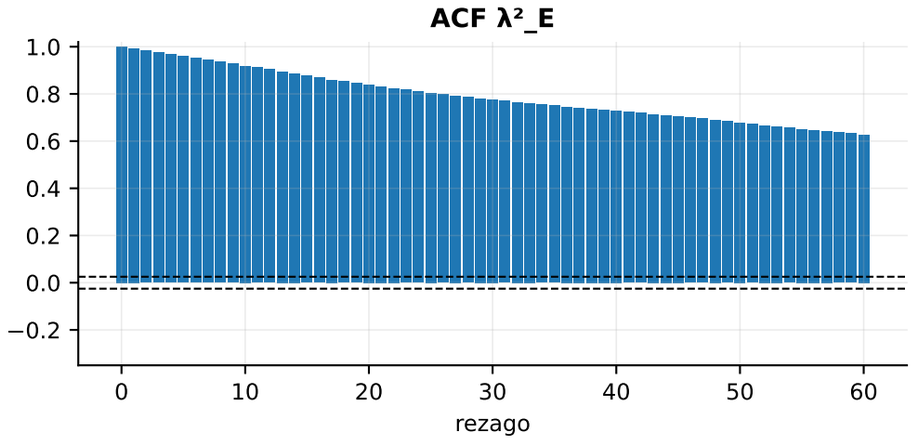


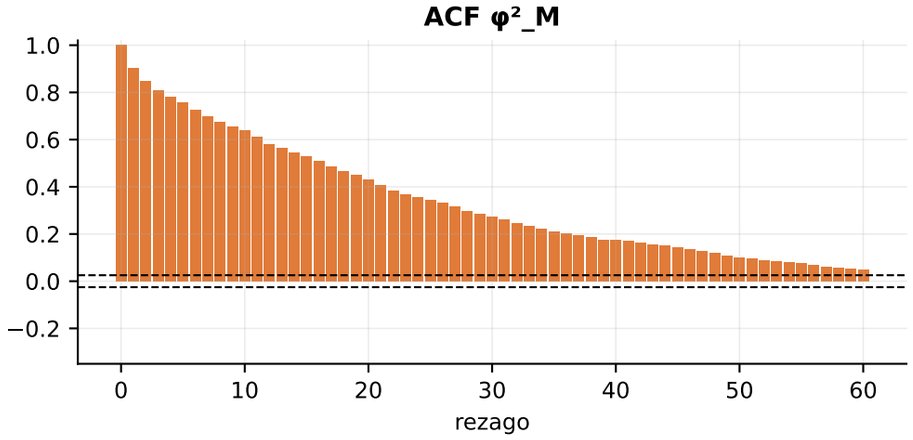
M4 — Convergencia III: autocorrelación de las cadenas

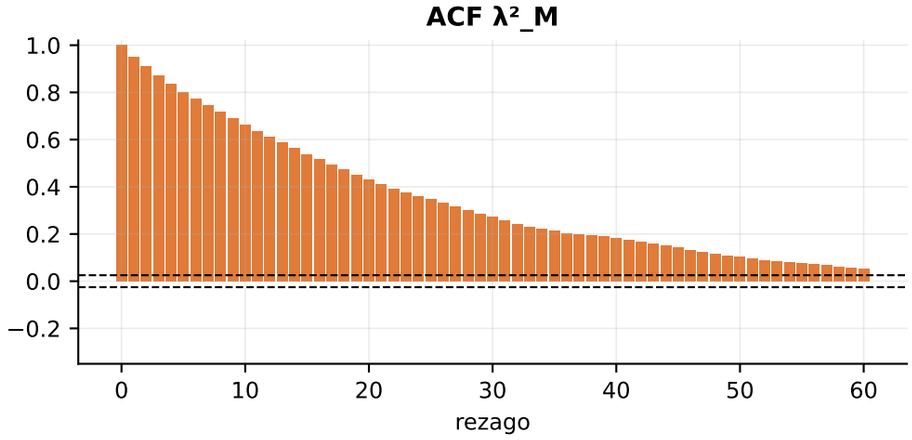


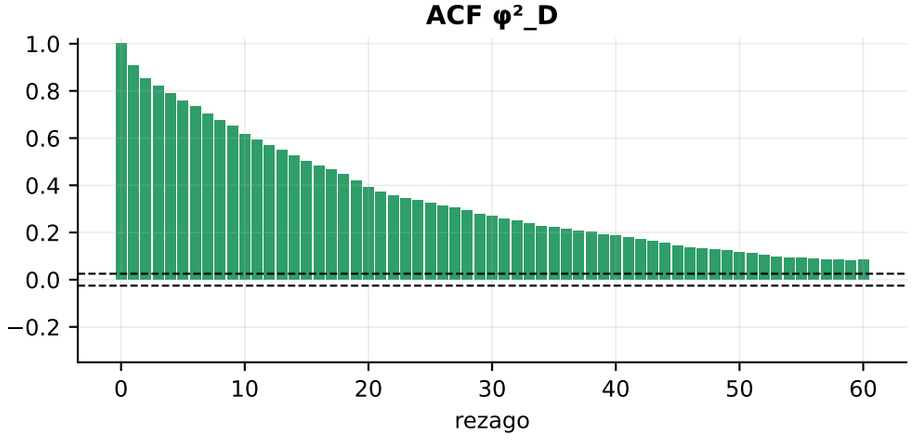


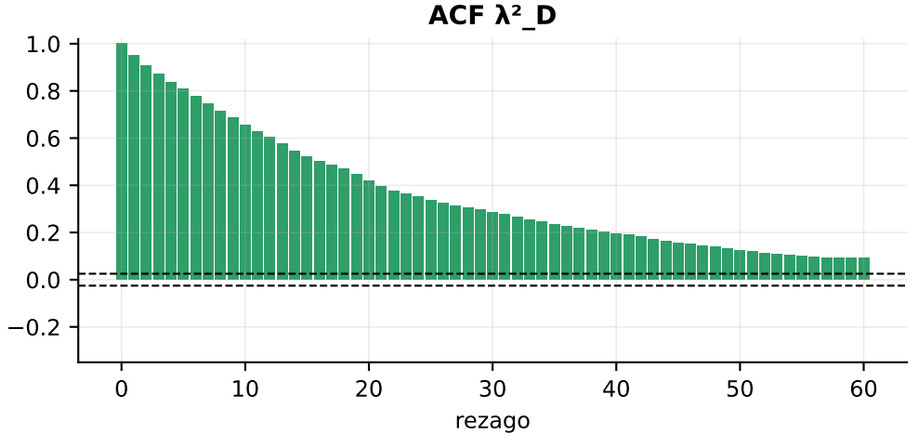






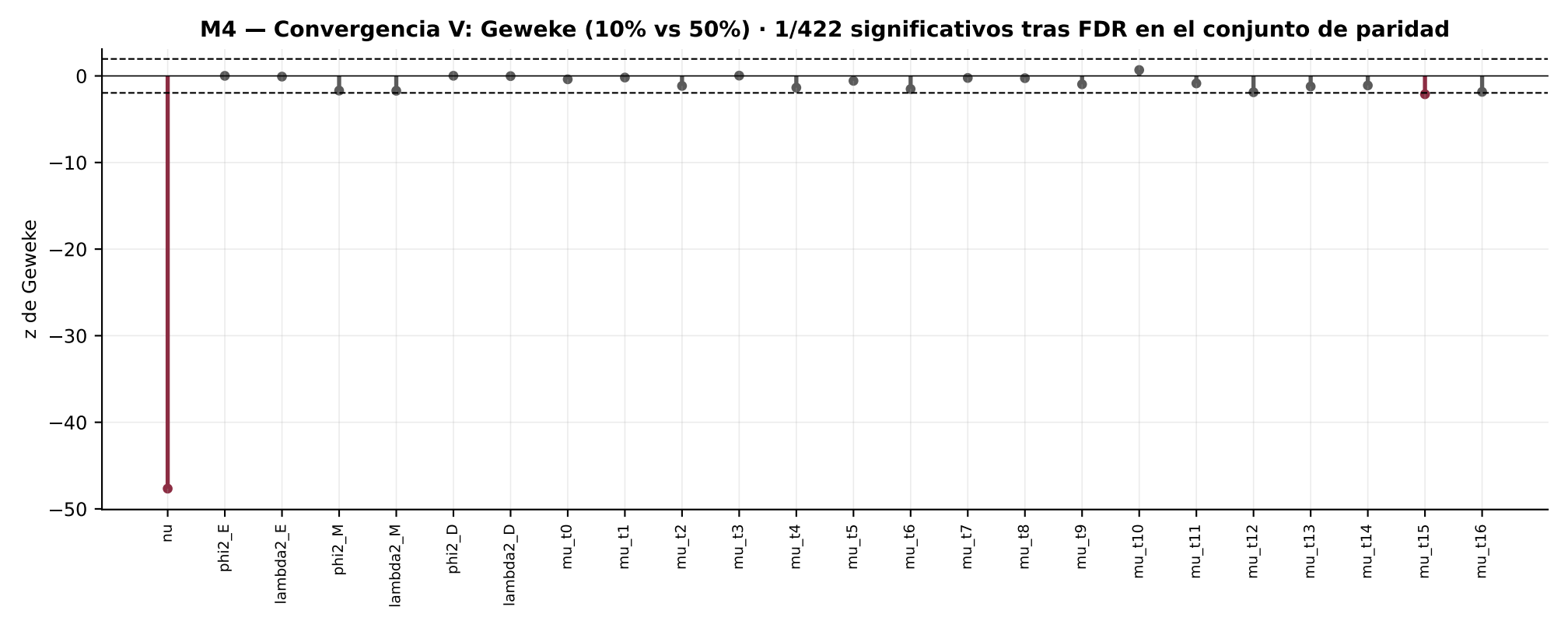


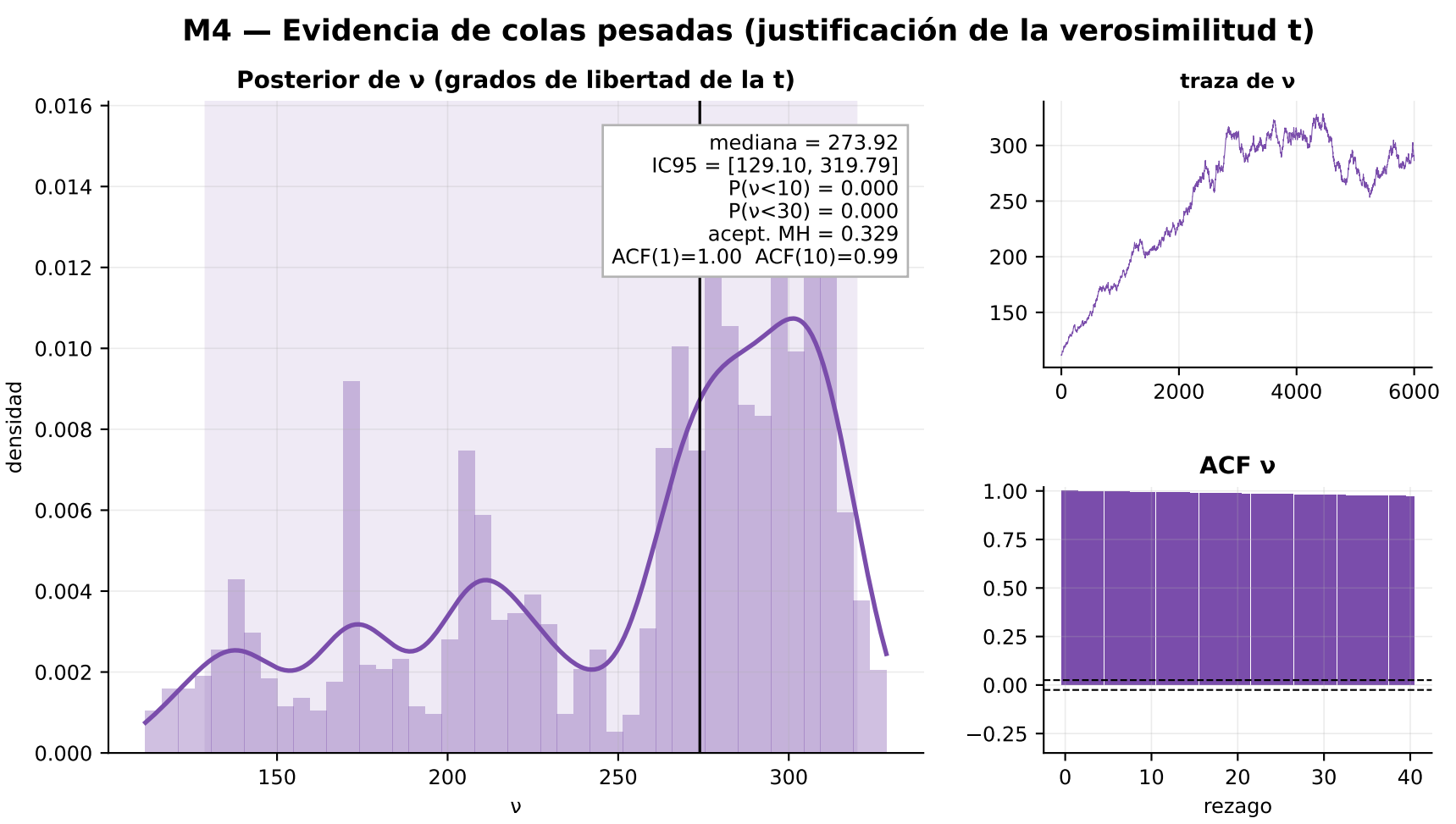


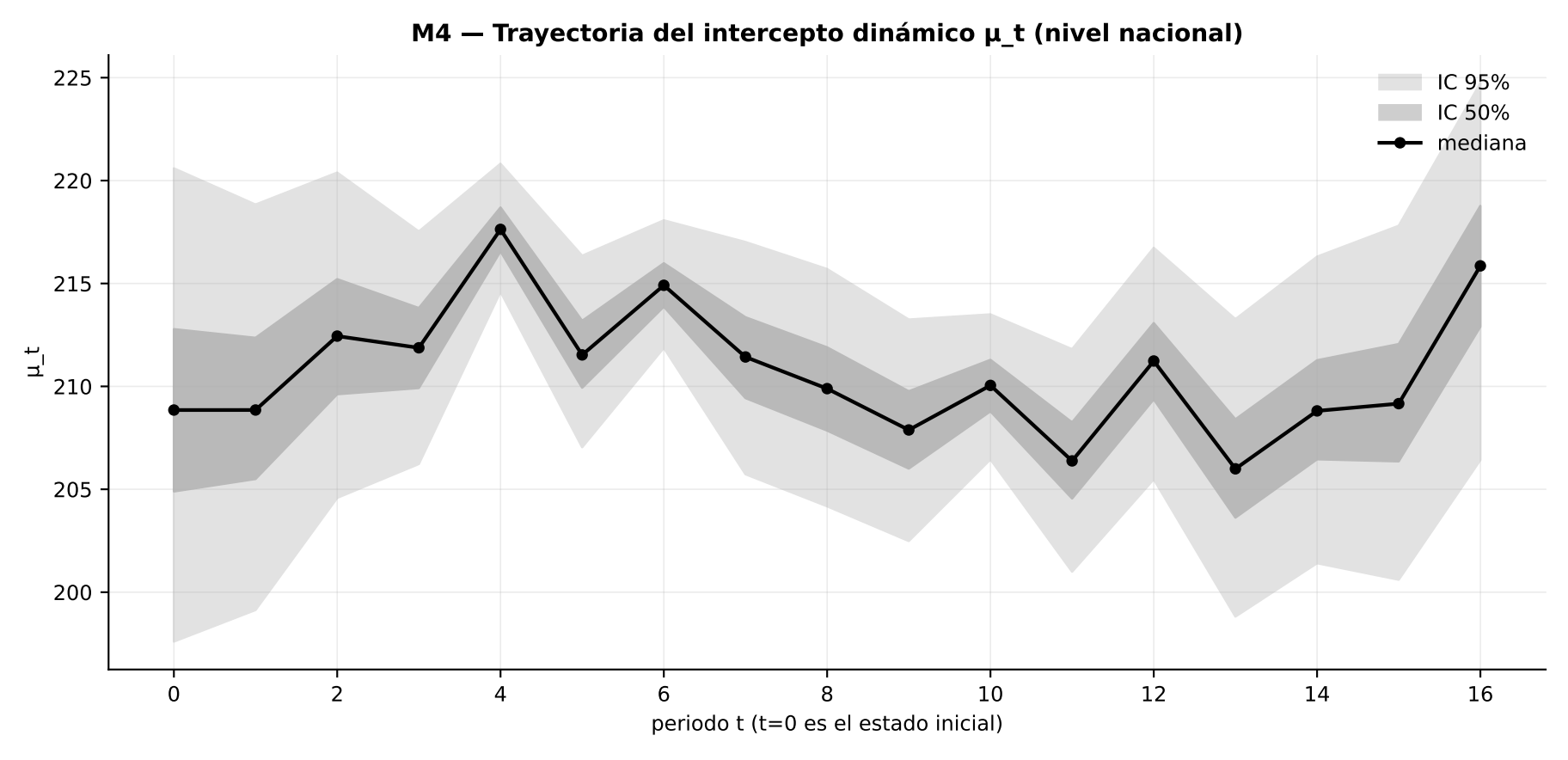


M4 — Convergencia IV: malla completa de θ (todos los coeficientes × t)

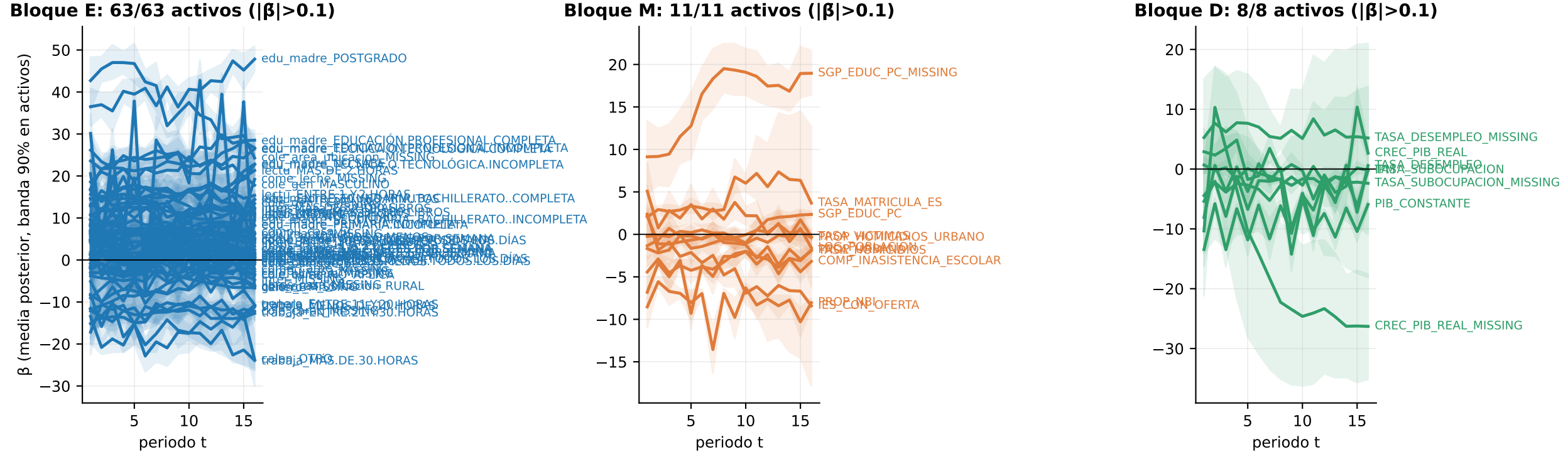



M4 — Convergencia V: Geweke (10% vs 50%) · 1/422 significativos tras FDR en el conjunto de paridad
z de Geweke
0
−10
−20
−30
−40
−50
nu
phi2_E
lambda2_E
phi2_M
lambda2_M
phi2_D
lambda2_D
mu_t0
mu_t1
mu_t2
mu_t3
mu_t4
mu_t5
mu_t6
mu_t7
mu_t8
mu_t9
mu_t10
mu_t11
mu_t12
mu_t13
mu_t14
mu_t15
mu_t16


M4 — Evidencia de colas pesadas (justificación de la verosimilitud t)
Posterior de ν (grados de libertad de la t)
mediana = 273.92
IC95 = [129.10, 319.79]
P(ν<10) = 0.000
P(ν<30) = 0.000
acept. MH = 0.329
ACF(1)=1.00 ACF(10)=0.99
densidad
ν
traza de ν
ACF ν
rezago

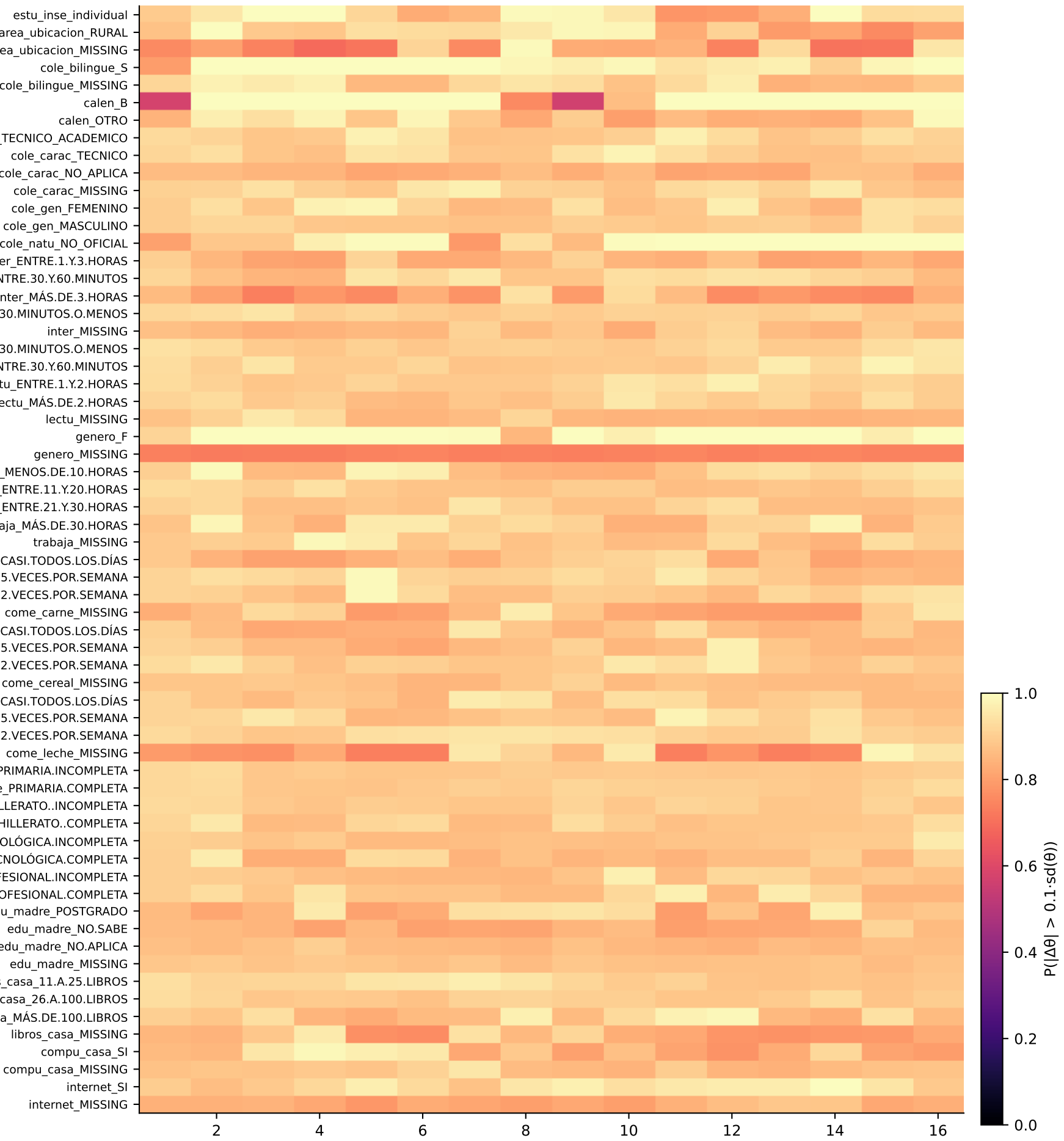

M4 — Trayectoria del intercepto dinámico μ_t (nivel nacional)
IC 95%
IC 50%
mediana
μ_t
225
220
215
210
205
200
0
2
4
6
8
10
12
14
16
periodo t (t=0 es el estado inicial)

M4 — Sendas dinámicas de los coeficientes por bloque

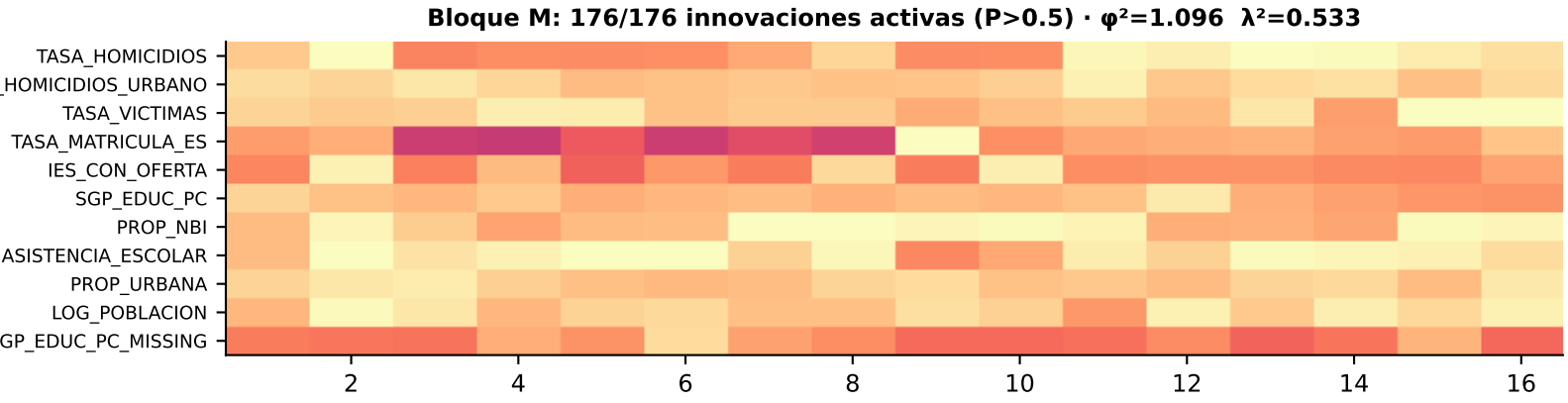


M4 — Shrinkage del LASSO dinámico sobre las innovaciones ω

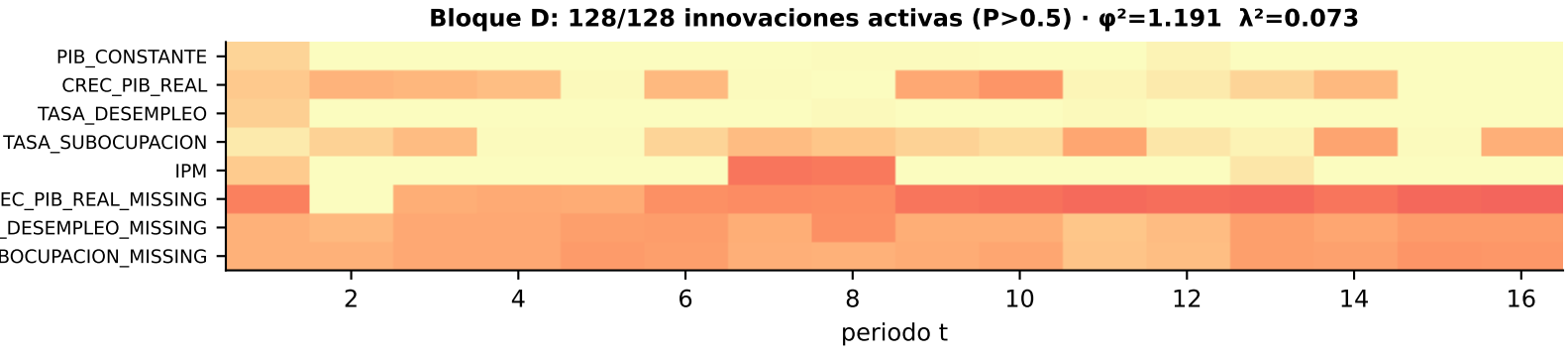
Bloque E: 1008/1008 innovaciones activas (P>0.5) · φ²=1.291 λ²=0.142



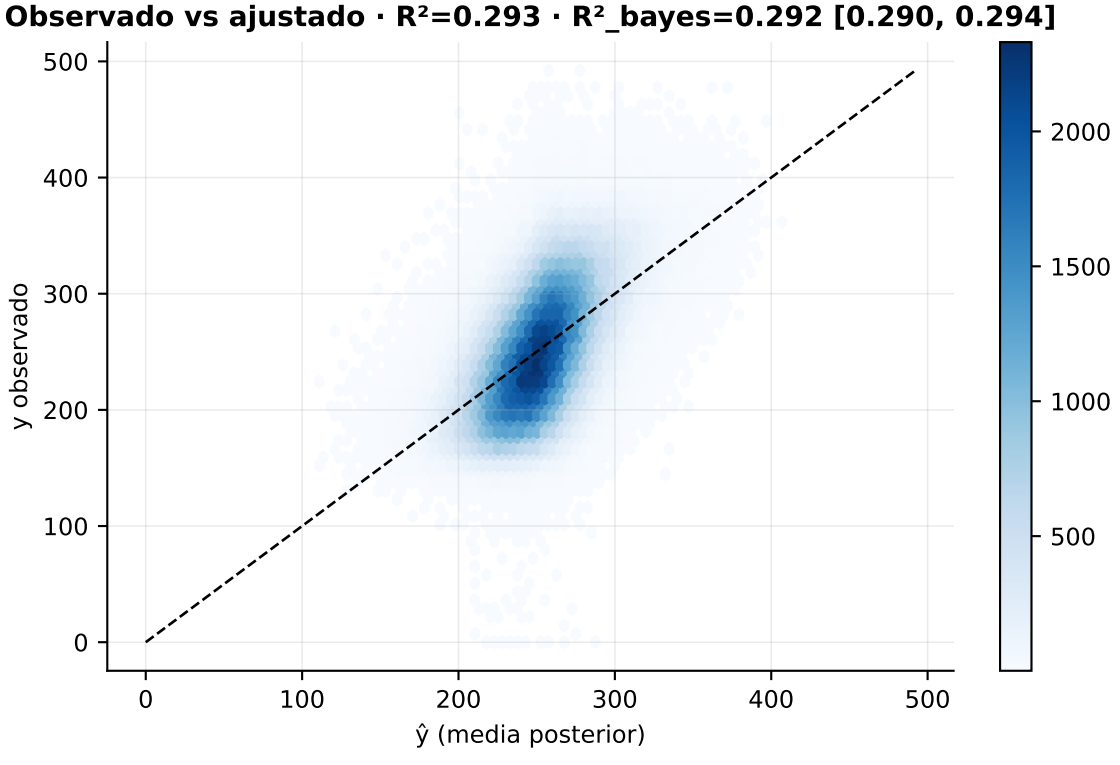
Bloque M: 176/176 innovaciones activas (P>0.5) · φ²=1.096 λ²=0.533

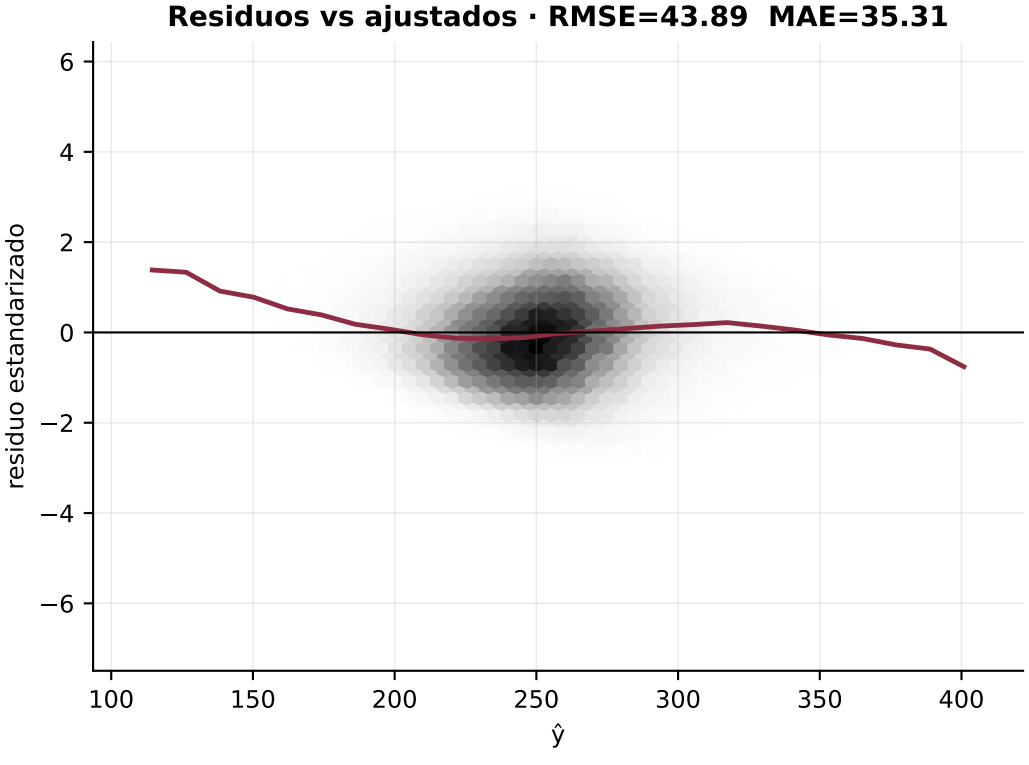


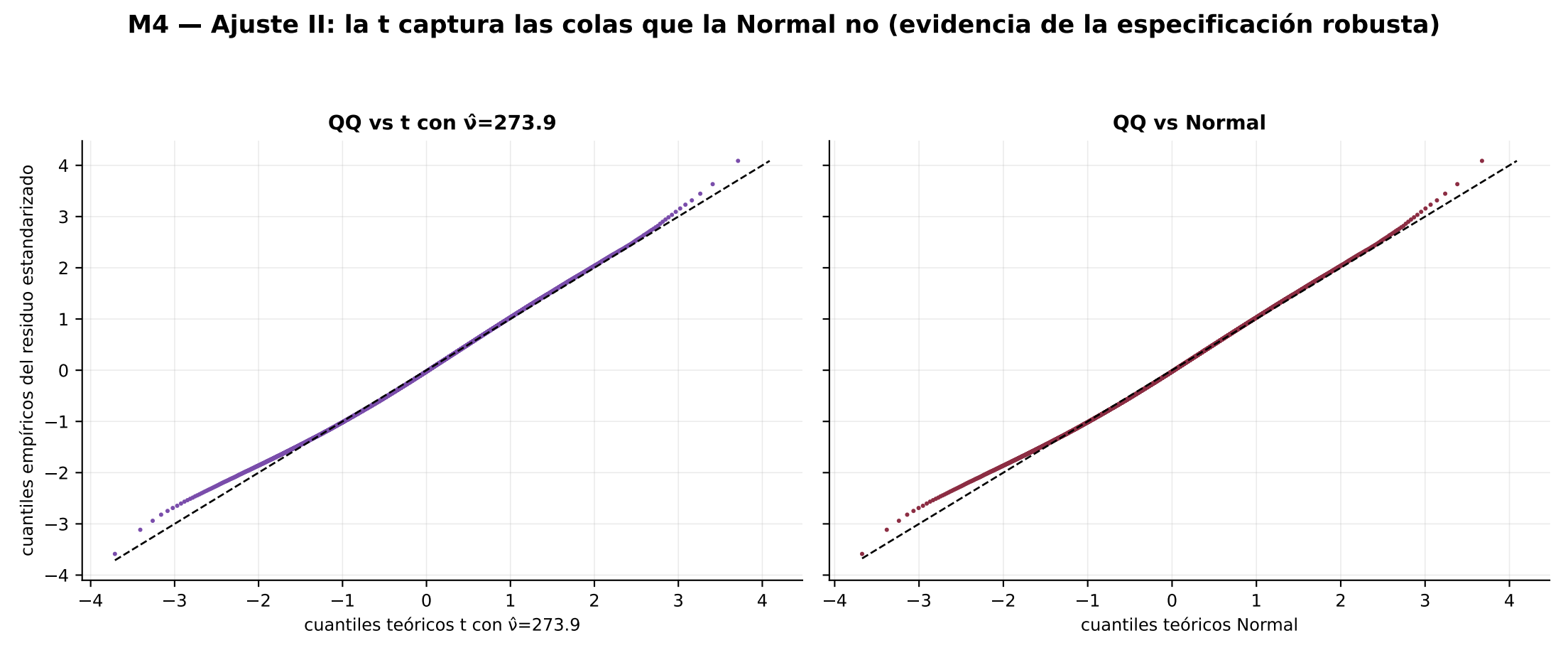
Bloque D: 128/128 innovaciones activas (P>0.5) · φ²=1.191 λ²=0.073

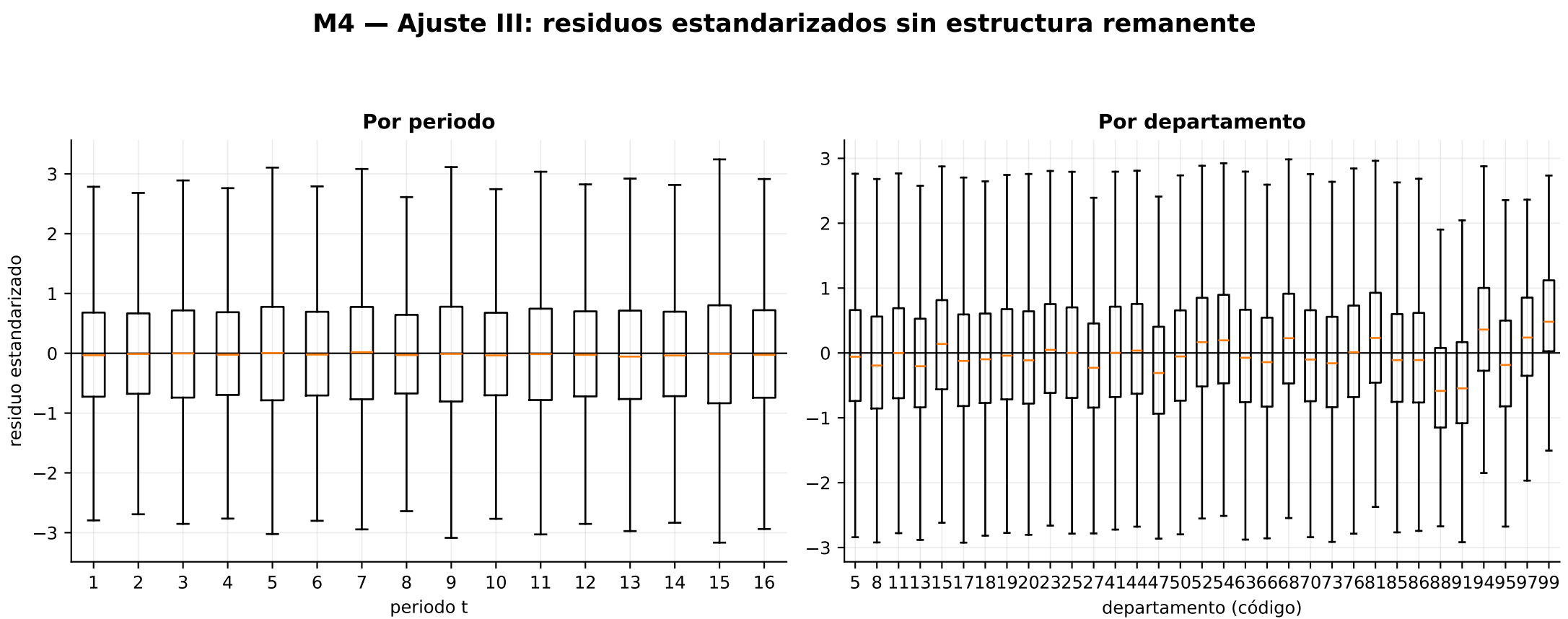


M4 — Ajuste I: capacidad explicativa y estructura de residuos

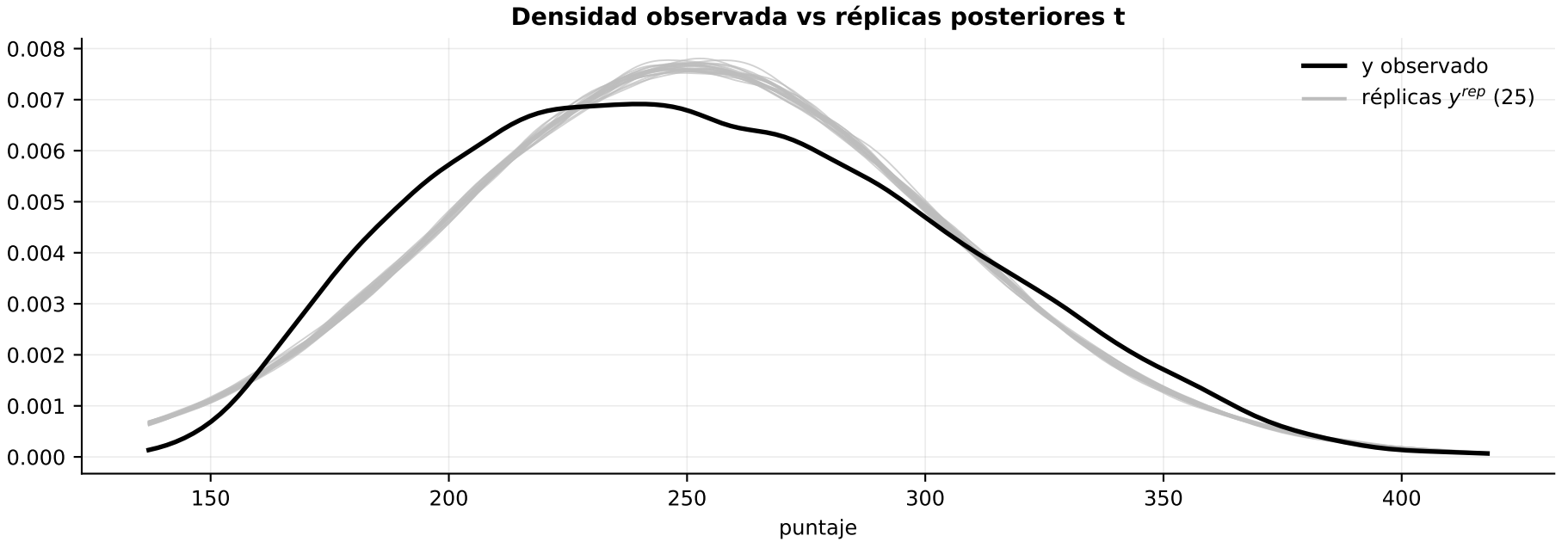


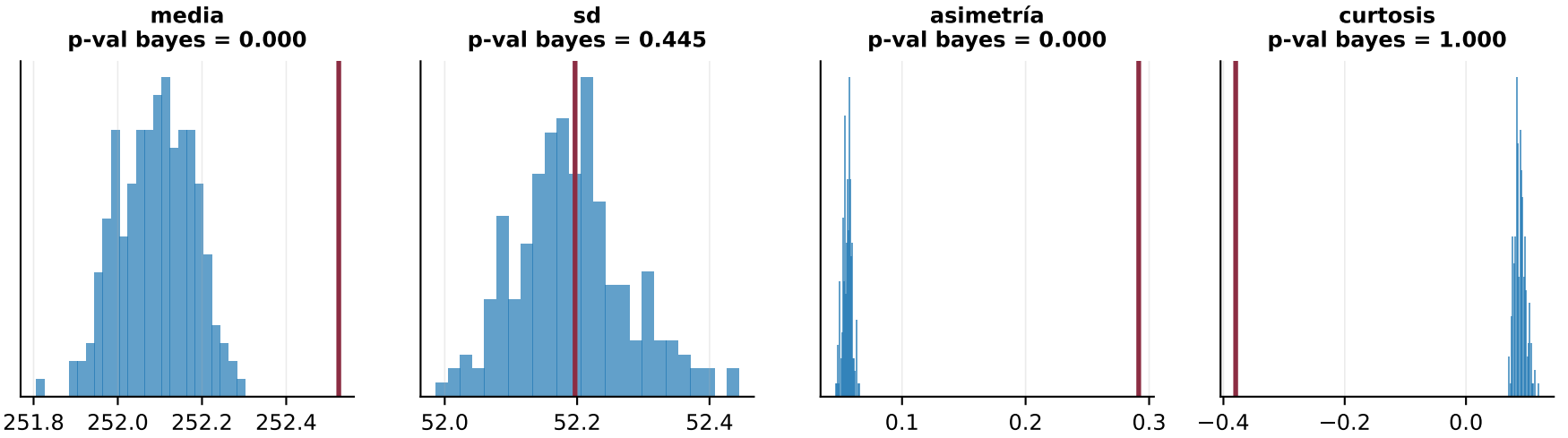



M4 — Ajuste II: la t captura las colas que la Normal no (evidencia de la especificación robusta)
QQ vs t con ν̂=273.9
QQ vs Normal
cuantiles empíricos del residuo estandarizado
cuantiles teóricos t con ν̂=273.9
cuantiles teóricos Normal
−4
−3
−2
−1
0
1
2
3
4

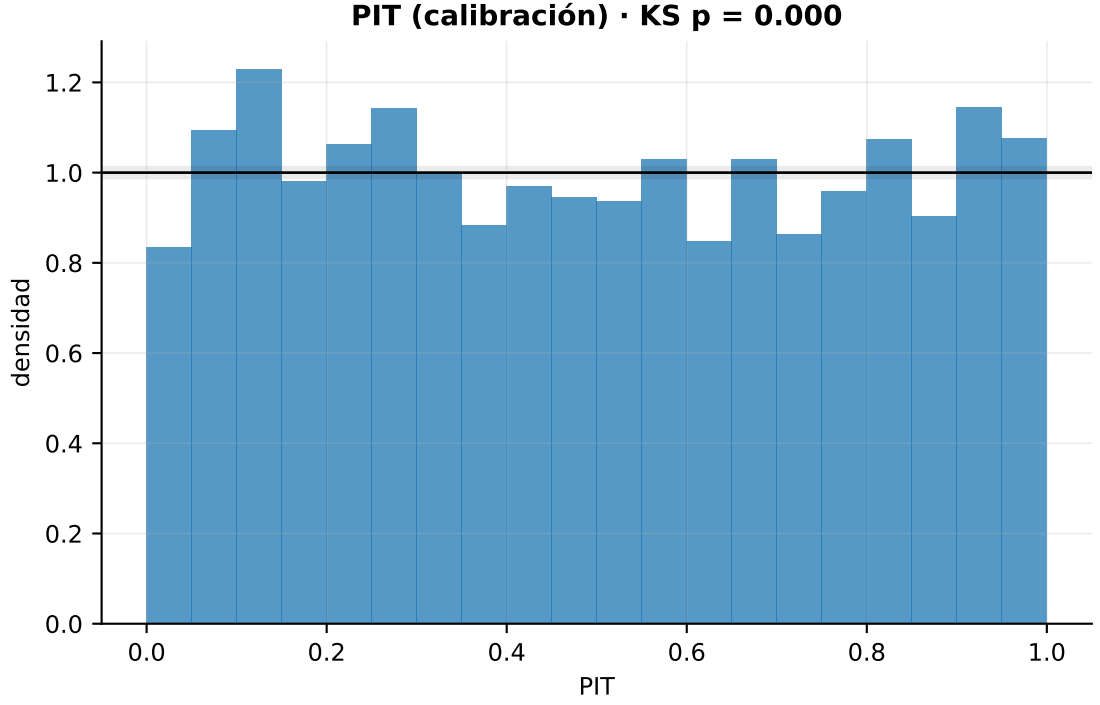

M4 — Ajuste III: residuos estandarizados sin estructura remanente
Por periodo
Por departamento
residuo estandarizado
periodo t
departamento (código)

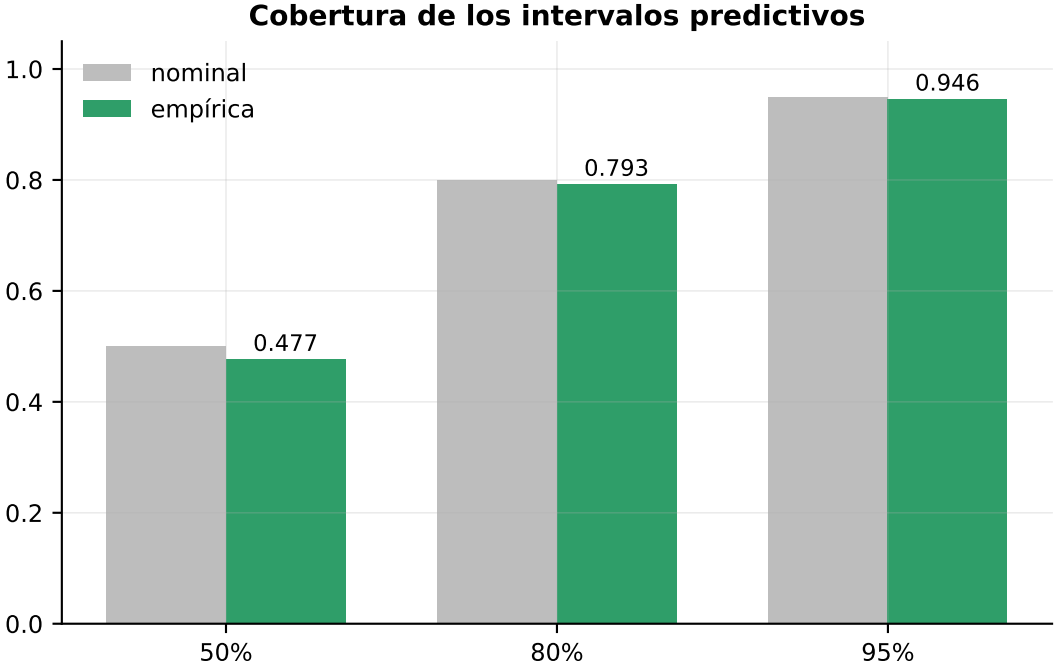
M4 — Validez predictiva I: posterior predictive checks





M4 — Validez predictiva II: calibración





M4 — Heterogeneidad de las escalas observacionales

